ISyE 3770 HW7

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2024-11-05

Problem 7.2.6

We begin by calculating the standard error of the sample mean:

$$SE = \frac{\sigma}{\sqrt(n)} = \frac{1.5}{\sqrt{49}} = 0.2143\ minutes$$

We then calculate the Z score:

$$Z = \frac{\bar{X} - \mu}{SE}$$

Part A

$$Z = \frac{10 - 8.2}{0.2143} = 8.4$$

We observe that 8.4 standard deviations above the mean is high and therefore, the probability of $Z = 8.4$ is approximately 1.

Thus, we have $P(\bar{X} < 10) \approx 1$.

Part B

$$Z_u = \frac{5 - 8.2}{0.2143} = -14.93$$
$$Z_l = \frac{10 - 8.2}{0.2143} = 8.4$$

Since $Z = -14.93$ is very low and $Z = 8.4$ is very high, the probability between these two scores is approximately one.

Therefore, we have: $P(5 < \bar{X} < 10) \approx 1$.

Part C

$$Z = \frac{6 - 8.2}{0.2143} = -10.27$$

Since $Z = -10.27$ is very low, the probability corresponding to this Z-Score is approximately zero.

Therefore, we have $P(\bar{X} < 6) \approx 0$.

7.3.7

Part A

We observe $\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$, with $n=24.

Therefore, we have the following:

$$\sum_{i=1}^{n} x_i = 10160$$

```
oxide_thickness <- c(425, 431, 416, 419, 421, 436, 418, 410, 431, 433,
                     423, 426, 410, 435, 436, 428, 411, 426, 409, 437,
                     422, 428, 413, 416)

sum(oxide_thickness)
```

```
## [1] 10160
```

$$\bar{X} = \frac{10160}{24} = 423.33 \ angstroms$$

Part B

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{X})^2}{n-1}}$$

```
oxide_thickness <- c(425, 431, 416, 419, 421, 436, 418, 410, 431, 433,
                     423, 426, 410, 435, 436, 428, 411, 426, 409, 437,
                     422, 428, 413, 416)

mean_oxide_thickness <- mean(oxide_thickness)
sum((oxide_thickness - mean_oxide_thickness)^2)
```

```
## [1] 1897.333
```

$$s = \sqrt{\frac{1897.33}{24-1}} = 9.082 \ angstroms$$

Part C

$$SE = \frac{s}{\sqrt{n}}$$

$$SE = \frac{9.082}{\sqrt{24}} = 1.854 \ angstroms$$

Part D

When we organize the set, we find the 12th value is 423 and the 13th value is 425. Therefore, we have the following:

$$median = \frac{423 + 425}{2} = 424 \ angstroms$$

Part E

We define p to be the following:

$$p = \frac{(\#\ wafers\ with\ thickness > 430)}{n}$$

Therefore, we have $p = \frac{7}{24} = 0.2917$.

Problem 7.3.9

Part A

We wish to show $E[\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2$.

Since $\bar{X}_1$ is the sample mean for population 1, we have $E[\bar{X}_1] = \mu_1$. Population 2 follows this same trend: $E[\bar{X}_2] = \mu_2$.

Therefore, we see the following:

$$E[\bar{X}_1 - \bar{X}_2] = E[\bar{X}_1] - E[\bar{X}_2] = \mu_1 - \mu_2$$

We find $\bar{X}_1 - \bar{X}_2$ is an unbiased estimator of $\mu_1 - \mu_2$.

Part B

We find the following:

$$Var[\bar{X}_1] = \frac{\sigma_1^2}{n_1}$$
$$Var[\bar{X}_2] = \frac{\sigma_2^2}{n_2}$$

We find the variance of the difference of two ind. estimators is the sum of their variances.

$$Var[\bar{X}_1 - \bar{X}_2] = Var[\bar{X}_1] + Var[\bar{X}_2] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

$$SE[\bar{X}_1 - \bar{X}_2] = \sqrt{Var[\bar{X}_1] + Var[\bar{X}_2]} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

We can estimate the standard error using the sample variances $S_1^2\ and\ S_2^2$ in place of $\sigma_1^2\ and\ \sigma_2^2$.

Part C

For each sample variance, we have the following:

$$E[S_i^2] = \sigma^2,\ for\ i = 1, 2$$

We further compute the expected value:

$$E[(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2] = (n_1 - 1)E[S_1^2] + (n_2 - 1)E[S_2^2]$$

With simple substitution, we have the following:

$$(n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2 = ((n_1 - 1) + (n_2 - 1))\sigma^2 = (n_1 + n_2 - 2)\sigma^2$$

Finally, we have:

$$E[S_p^2] = \frac{E[n_1 - 1)S_1^2 + (n_2 - 1)S_2^2]}{n_1 + n_2 - 2} = \frac{(n_1 + n_2 - 2)\sigma^2}{n_1 + n_2 - 2} = \sigma^2$$

Therefore, we find S_p^2 is an unbiased estimator of σ^2 when $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

Problem 7.3.11

Part A

We wish to show $E[\frac{X_1}{n_1} - \frac{X_2}{n_2}] = p_1 - p_2$.

Since X_1 *and* X_2 follows a binomial distribution, we have the following:

$$E[\frac{X_1}{n_1}] = \frac{E[X_1]}{n_1} = \frac{n_1 p_1}{n_1} = p_1$$

$$E[\frac{X_2}{n_2}] = \frac{E[X_2]}{n_2} = \frac{n_2 p_2}{n_2} = p_2$$

We compute the following:

$$E[\frac{X_1}{n_1} - \frac{X_2}{n_2}] = E[\frac{X_1}{n_1}] - E[\frac{X_2}{n_2}] = p_1 - p_2$$

Therefore, we find $\frac{X_1}{n_1} - \frac{X_2}{n_2}$ is an unbiased estimator of $p_1 - p_2$.

Part B

$$Var[\hat{p_1}] = \frac{p_1(1-p_1)}{n_1}$$

$$Var[\hat{p_2}] = \frac{p_2(1-p_2)}{n_2}$$

We take the variance of the difference to have:

$$Var[\hat{p_1} - \hat{p_2}] = Var[\hat{p_1}] + Var[\hat{p_2}] = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$$

Therefore, we have a standard error of:

$$SE = \sqrt{Var[\hat{p_1} - \hat{p_2}]} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Part C

We replace p_1 and p_2 with $\hat{p_1}$ and $\hat{p_2}$ which are the sample proportions to result in:

$$SE_{estimate} = \sqrt{\frac{\hat{p_1}(1-\hat{p_1})}{n_1} + \frac{\hat{p_2}(1-\hat{p_2})}{n_2}}$$

It is important we define $\hat{p_i} = \frac{X_i}{n_i}$.

Part D

We compute the sample proportions:

ASU: $\hat{p_1} = \frac{X_1}{n_1} = \frac{150}{200} = 0.75$

VT: $\hat{p_2} = \frac{X_2}{n_2} = \frac{185}{250} = 0.74$

Therefore, an estimate of $p_1 - p_2$ is the following:

$$\hat{p_1} - \hat{p_2} = 0.75 - 0.74 = 0.01$$

Part E

We compute each variance component:

$$Var[\hat{p_1}] = \frac{0.75(0.25)}{200} = 0.0009375$$
$$Var[\hat{p_2}] = \frac{0.74(0.26)}{250} = 0.0007696$$

Therefore, we have:
$$Var[\hat{p_1} - \hat{p_2}] = 0.0009375 + 0.0007696 = 0.0017071$$

Which means the standard error is $\sqrt{0.0017071} = 0.04132$.

Problem 7.4.1

We define $L(p)$ as the following:

$$L(p) = \prod_{i=1}^{n} f(X_i) = \prod_{i=1}^{n} p(1-p)^{X_i - 1}$$

Simplifying the product:

$$L(p) = p^n \times \prod_{i=1}^{n} (1-p)^{X_i - 1} = p^n (1-p)^{\sum_{i=1}^{n}(X_i - 1)} = p^n (1-p)^{S-n}$$

where $S = \sum_{i=1}^{n} X_i$.

We now take the natural log:

$$\ell(p) = \ln L(p) = n \ln p + (S - n)\ln(1-p)$$

Now, we take the derivative with respect to p:

$$\frac{d\ell(p)}{dp} = \frac{n}{p} - \frac{S-n}{1-p}$$

Finally, we set the derivative equal to zero:

$$\frac{n}{p} - \frac{S-n}{1-p} = 0$$
$$n(1-p) - (S-n)p = 0$$
$$n - np - Sp + np = n - Sp = 0$$
$$n = Sp \quad \Rightarrow \quad \hat{p} = \frac{n}{S}$$

Therefore, we arrive at our final answer of:

$$\hat{p} = \frac{n}{\sum_{i=1}^{n} X_i}$$

Question Two

Part A

For a normal distribution, we have:

$$\mathrm{E}[\bar{X}_n] = \mu, \quad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

$$\mathrm{E}[\bar{Y}_m] = \mu, \quad \mathrm{Var}(\bar{Y}_m) = \frac{\sigma^2}{m}$$

We compute $\mathrm{E}[\hat{\mu}_1]$:

$$\mathrm{E}[\hat{\mu}_1] = \mathrm{E}\left[\frac{\bar{X}_n + \bar{Y}_m}{2}\right] = \frac{\mathrm{E}[\bar{X}_n] + \mathrm{E}[\bar{Y}_m]}{2} = \frac{\mu + \mu}{2} = \mu$$

Similarly, we now compute $\mathrm{E}[\hat{\mu}_2]$:

$$\mathrm{E}[\hat{\mu}_2] = \mathrm{E}\left[\frac{n\bar{X}_n + m\bar{Y}_m}{n+m}\right] = \frac{n\mathrm{E}[\bar{X}_n] + m\mathrm{E}[\bar{Y}_m]}{n+m} = \frac{n\mu + m\mu}{n+m} = \mu$$

Therefore, we find both estimators $\hat{\mu}_1$ and $\hat{\mu}_2$ are unbiased estimators of μ.

Part B

We first calculate the variance of $\hat{\mu}_1$. Since $\bar{X}_n$ and $\bar{Y}_m$ are independent:

$$\mathrm{Var}(\hat{\mu}_1) = \mathrm{Var}\left(\frac{\bar{X}_n + \bar{Y}_m}{2}\right) = \left(\frac{1}{2}\right)^2 \left[\mathrm{Var}(\bar{X}_n) + \mathrm{Var}(\bar{Y}_m)\right] = \frac{1}{4}\left(\frac{\sigma^2}{n} + \frac{\sigma^2}{m}\right)$$

$$\mathrm{Var}(\hat{\mu}_1) = \frac{\sigma^2}{4}\left(\frac{1}{n} + \frac{1}{m}\right).$$

Now, we find the variance of $\hat{\mu}_2$:

$$\hat{\mu}_2 = \frac{n}{n+m}\bar{X}_n + \frac{m}{n+m}\bar{Y}_m$$

Let $a = \dfrac{n}{n+m}$ and $b = \dfrac{m}{n+m}$.

Since $\bar{X}_n$ and $\bar{Y}_m$ are independent:

$$\mathrm{Var}(\hat{\mu}_2) = a^2\mathrm{Var}(\bar{X}_n) + b^2\mathrm{Var}(\bar{Y}_m) = \left(\frac{n}{n+m}\right)^2 \frac{\sigma^2}{n} + \left(\frac{m}{n+m}\right)^2 \frac{\sigma^2}{m}$$

$$\mathrm{Var}(\hat{\mu}_2) = \frac{n^2\sigma^2}{(n+m)^2 n} + \frac{m^2\sigma^2}{(n+m)^2 m} = \frac{n\sigma^2}{(n+m)^2} + \frac{m\sigma^2}{(n+m)^2}$$

$$\mathrm{Var}(\hat{\mu}_2) = \frac{\sigma^2}{(n+m)^2}(n+m) = \frac{\sigma^2}{n+m}$$

Therefore, we find the following:

$$\mathrm{Var}(\hat{\mu}_1) = \frac{\sigma^2}{4}\left(\frac{1}{n} + \frac{1}{m}\right), \quad \mathrm{Var}(\hat{\mu}_2) = \frac{\sigma^2}{n+m}$$

Part C

The relative efficiency of $\hat{\mu}_2$ to $\hat{\mu}_1$ is defined as:

$$\text{RE} = \frac{\text{Var}(\hat{\mu}_1)}{\text{Var}(\hat{\mu}_2)}$$

$$\text{Var}(\hat{\mu}_1) = \frac{\sigma^2}{4}\left(\frac{1}{n} + \frac{1}{m}\right) = \frac{\sigma^2}{4}\left(\frac{m+n}{nm}\right)$$

$$\text{Var}(\hat{\mu}_2) = \frac{\sigma^2}{n+m}$$

Now, we compute the relative frequency:

$$\text{RE} = \frac{\text{Var}(\hat{\mu}_1)}{\text{Var}(\hat{\mu}_2)} = \frac{\frac{\sigma^2}{4}\left(\frac{m+n}{nm}\right)}{\frac{\sigma^2}{n+m}} = \frac{\frac{\sigma^2(n+m)}{4nm}}{\frac{\sigma^2}{n+m}} = \frac{(n+m)^2}{4nm}$$

Therefore, we arrive at our final answer of:

$$\text{Relative Efficiency} = \frac{4nm}{(n+m)^2}$$

Part D

We find that an estimator with a smaller variance has greater efficiency since:

$$\text{RE} = \frac{\text{Var}(\hat{\mu}_2)}{\text{Var}(\hat{\mu}_1)} = \frac{4nm}{(n+m)^2}$$

We observe $4nm \leq (n+m)^2$ for all positive integers. Therefore:

$$(n+m)^2 \geq 4nm \implies \frac{4nm}{(n+m)^2} \leq 1$$

We can simply observe $\text{RE} \leq 1$, meaning the following:

$$\text{Var}(\hat{\mu}_2) \leq \text{Var}(\hat{\mu}_1).$$

Finally, we arrive at our final conclusion of the estimator $\hat{\mu}_2$ has a smaller variance and is therefore more efficient than $\hat{\mu}_1$ based on relative efficiency.